

University of Botswana

Department of Mathematics

Tutorial 3: Differentiation

Part I: Limit Process or Definition of the Derivative

In question 1–4, find the derivative of the following functions by the limit process (definition of the derivative).

1. $f(x) = x^2 - 5x + 3$
2. $f(x) = \frac{1}{x-1}$
3. $f(x) = \sqrt{x-2}$
4. Prove from first principles that $\frac{d}{dx}[\cos x] = -\sin x$.

Part II: Basic rules of differentiation

In question 5–7, find the derivative of the following functions.

5. $f(x) = \frac{1}{4}x^4 + 5x^3 + 20 - 2x + 3x^{-3} + e^x$
6. $f(x) = \frac{3}{x^2} - \ln x + 6\sqrt{x} + 5\sin x$
7. $h(x) = \frac{4x^3+2x+5}{2x}$

Part III: Rates of change

8. Find the equation of the tangent line to $f(x) = x^3 - 2x$ at $x = -1$.
9. Given that $f(1) = -3$ and $f'(1) = 6$ the equation of the tangent line to the graph of $y = f(x)$ at $x = 1$.
10. A ball is thrown upward from ground level with initial velocity 64 m/s. Its height is given by $s(t) = -16t^2 + 64t$. Find:
 - (a) The velocity function.
 - (b) The time when the ball reaches its maximum height.
 - (c) The velocity at that time.

Part IV: Product and Quotient Rules

In Exercises 11–16, use the Product or Quotient Rule where applicable to find the first derivative or $\frac{dy}{dx}$.

11. $y = 6x^2 \cos x$

12. $y = (3x^2 - 4\sqrt{x})(x^3 + 5)$

13. $h(t) = (t^{\frac{2}{5}} - t^2) \sec t$

14. $g(x) = \frac{x^2}{2\sqrt{x+1}}$

15. $f(t) = \frac{3 - \frac{1}{t}}{t+1}$

16. Prove by quotient rule that $\frac{d}{dx}[\tan x] = \sec^2 x$.

Part V: Higher Order Derivatives

For Question 17–19, find the given higher order derivative.

17. $f(x) = x^2 - \cos x, \quad f'''(x)$

18. $f'(x) = \frac{1}{2}x^4 - \frac{2}{x}, \quad f^{(4)}(x)$

19. $f^{(4)}(x) = 2\sqrt{x}, \quad f^{(5)}(x)$

Part VI: Chain Rule

In Exercises 20–25, find the derivative by using chain rule

20. $g(x) = 3(4 - 9x)^4$

21. $f(x) = \sqrt[3]{12x - 5 \cos x}$

22. $f(t) = -\frac{3}{(t-2)^4}$

23. $f(x) = \sin(2x + 3)$

24. $f(x) = -5 \cos(\pi x^2)$

25. $f(\theta) = \sin^2 3\theta$